

An Exact Solution to the Drop-the-Handkerchief Timing Game

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Abstract

Drop the Handkerchief (DTH) is a fully observed, two-player, zero-sum stochastic game with simultaneous exact-second actions, a hard 300-second capacity, and a probabilistic revival rule. We solve the complete game. Once a player can no longer survive an injection, the game never reads their cumulative time dead again. Quotienting by that observation collapses 5.27×10^9 reachable states to 289,374,121 equivalence classes, a factor of 18.2. A potential function organizes those classes into 1,201 layers and permits a single backward sweep without search or reachability analysis. Every stored value is accepted only with an explicit equilibrium certificate whose saddle gap against the full 60×60 stage matrix is at most 10^{-6} . A ladder of certified solvers produced these values; 99.8% of the equilibria had full support. The production sweep ran once through a parallel Rust kernel, while a Python reference matched it byte for byte on every region computed by both. The certified root value is

$$V(0, 0, 0, 0) = 0.0898500728095116,$$

so under optimal play the first Dropper wins with probability ≈ 0.5449 . The computation took 5,432 seconds, or about 1.51 hours, on one twelve-core desktop, against a previously recorded single-core projection of 3.7–5.1 years.

1 Introduction

DTH is the pure timing core of the “Surpassing the Leader” wager from the manga *Usogui*. Two players alternate the roles of *Dropper* and *Checker*. Each round both simultaneously commit to a second in $\{1, \dots, 60\}$; the check succeeds if it does not precede the drop, in which case the checker absorbs the inclusive elapsed interval, also called squandered time (ST), as poison in a personal 300-second cylinder, and the roles swap. A failed check injects the checker with their accumulated load plus a fixed 60-second penalty; the injection is fatal unless a frozen, monotone revival law (Figure 1) saves them, in which case their cylinder resets and their cumulative time dead (TTD) grows. Overflowing the cylinder is fatal outright. The last player alive wins. That is the entire game.

The game is finite but large, stochastic, and simultaneous. Its stage games require mixed strategies, and its state space was, until now, handled by bounded search around certified anchor values. This paper records the complete solution and the mathematics that made it cheap:

*Code and instructions to reproduce the results are available in the project repository.

1. a *per-player quotient* (Section 4): death eligibility is absorbing and zeroes the probability model, the only expression that ever reads a player’s time dead, so that coordinate collapses to a sentinel independently for each player;
2. a *potential* (Section 5) that strictly increases along every live transition, so a single descending sweep sees every child before its parents, thus eliminating search, reachability analysis, and frontier bookkeeping in one stroke;
3. a *certificate discipline* (Section 6): no value enters the table without a mixed-strategy pair whose one-step exploitability against the fully reconstructed stage matrix is at most $\tau_0 = 10^{-6}$, no matter which solver produced it;
4. the 61-class *Toeplitz structure* of the stage matrices (Section 3), which makes certificates and pure-saddle tests $O(60)$ or $O(60k)$ instead of $O(60^2)$.

Section 7 reports the computation, its cross-implementation audit, and the error analysis. Section 8 states what is and is not established.

2 The game

Definition 2.1 (State). *A live state is $x = (s_c, t_c, s_d, t_d) \in \mathcal{X}$, written from the perspective of the current Dropper: $s_c \in \{0, \dots, 299\}$ and $t_c \in \{0, \dots, 300\}$ are the current Checker’s accumulated load (“squandered time”, ST) and cumulative time dead (TTD), and (s_d, t_d) are the Dropper’s. Terminal payoffs are +1 when the current Dropper wins and −1 when they lose.*

Definition 2.2 (Actions and timing). *Both players simultaneously choose seconds $d, c \in \{1, \dots, 60\}$ (drop and check). The check succeeds iff $d \leq c$, and then the checker’s load grows by the inclusive interval $\ell = c - d + 1 \in \{1, \dots, 60\}$; equal seconds cost one second, not zero. If $s_c + \ell \geq 300$ the cylinder overflows and the checker dies: the transition ends with payoff +1 for the Dropper. Otherwise play continues from*

$$(s_c, t_c, s_d, t_d) \xrightarrow{\text{success, lag } \ell} (s_d, t_d, s_c + \ell, t_c),$$

with the roles swapped (the previous Checker drops next).

Definition 2.3 (Failure, dose, and revival). *On a failed check ($d > c$) the checker is injected with the dose $q = s_c + 60$. Survival is possible iff*

$$q < 300 \wedge q + t_c \leq 300 \quad \Longleftrightarrow \quad s_c \leq 239 \wedge s_c + t_c \leq 240, \quad (1)$$

where the second inequality is deliberately non-strict: a resulting cumulative TTD of exactly 300 remains eligible and thus survivable. When (1) holds, the checker revives with the frozen probability

$$p(s_c, t_c) = 0.95 \left(1 - \frac{s_c}{240}\right) 0.75^{t_c/60}, \quad (2)$$

their cylinder resets, and their TTD grows by the dose:

$$(s_c, t_c, s_d, t_d) \xrightarrow{\text{failure, revive}} (s_d, t_d, 0, t_c + s_c + 60) \quad \text{with probability } p(s_c, t_c),$$

else the transition ends with payoff +1. When (1) fails, the injection is fatal with certainty and $p \equiv 0$.

Revival probability on the survivable region

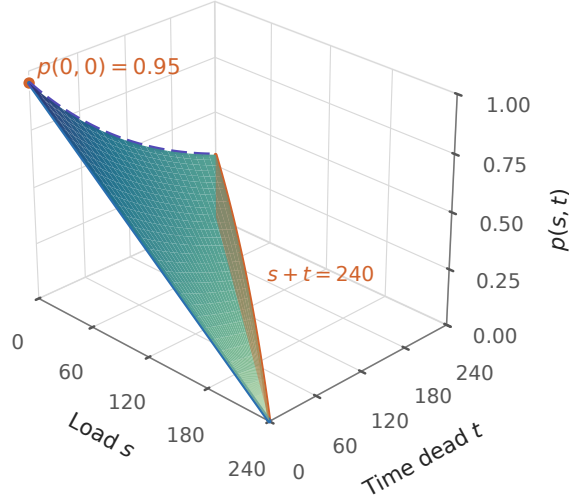


Figure 1: The frozen revival law (2). Color and height both encode probability. The blue edge isolates the linear loss from load; the purple edge isolates the exponential loss from time dead. The orange curtain marks $s + t = 240$: probability is positive on the curtain and drops to zero immediately beyond it.

The source gives no numerical revival odds, but it fixes the qualitative geometry: a failed check injects the vial contents plus sixty seconds against a five-minute cap; nearby deaths lower revival prospects enough to change strategy; and the referee’s condition separately affects resuscitation [1]. These facts motivate the separable form in (2). Since $q = s + 60$, the acute factor $1 - s/240 = (300 - q)/240$ is the simplest monotone path from a bare injection to the lethal dose, so each second of ST consumes an equal share of acute headroom. TTD records damage carried through earlier revivals; $0.75^{t/60}$ makes each additional death-minute retain a fixed fraction of the chance remaining after the acute dose. The source supports the signs and endpoint, not linearity, multiplicative separation, or the constants 0.95 and 0.75; those are frozen modeling choices beneath every result below.

Values are always from the current Dropper’s perspective and negate across the role swap; the game is zero-sum with terminal-only payoff in $\{-1, +1\}$, so all values lie in $[-1, 1]$. We call the checker profile (s, t) *dead* when (1) fails — the player is still playing, but any future failed check is certainly fatal for them. Figure 1 plots the law. It is smooth and positive on the alive triangle $s + t \leq 240$, then identically zero outside it. TTD is read nowhere else in the rules, so Section 4 collapses the zero region.

3 Finiteness and stage structure

Lemma 3.1 (Damage strictly increases). *Let $R(x) = s_c + t_c + s_d + t_d$. Every live transition strictly increases R : a successful lag ℓ adds $\ell \geq 1$, and a revived failure adds exactly 60 (the dose $s_c + 60$ enters TTD while the load s_c resets).*

Corollary 3.2 (The game is finite). *$R \leq 299 + 300 + 299 + 300 = 1198$ on live states, so every play*

reaches a terminal outcome within 1198 live transitions. The game therefore has a unique value $V : \mathcal{X} \rightarrow [-1, 1]$, obtained by backward induction over the finite acyclic transition graph [2], solving at each state the zero-sum simultaneous stage game by von Neumann’s minimax theorem [3]; the construction is the finite, absorbing case of a Shapley stochastic game [4].

Proposition 3.3 (The 61 transition classes). *Fix a live state x and let V assign values to its children. Define*

$$S_\ell = \begin{cases} +1, & s_c + \ell \geq 300, \\ -V(s_d, t_d, s_c + \ell, t_c), & \text{else,} \end{cases}$$

$$F = \begin{cases} +1, & (s_c, t_c) \text{ dead,} \\ p(-V(s_d, t_d, 0, t_c + s_c + 60)) + (1 - p), & \text{else.} \end{cases}$$

Then the 60×60 stage matrix of x , with the Dropper as row player, is

$$M(x)_{d,c} = \begin{cases} S_{c-d+1}, & c \geq d, \\ F, & c < d, \end{cases} \quad (3)$$

and $V(x) = \text{val}(M(x))$. In particular $M(x)$ carries only 61 distinct entries: it is Toeplitz on and above the diagonal and constant below it.

Proof. A successful cell depends on the actions only through the lag $\ell = c - d + 1$, and every failed cell has the identical transition distribution; the entries are the expected continuation values, negated for the role swap. The Bellman equation is Corollary 3.2. $\square$

Lemma 3.4 ($O(60)$ saddle reductions). *Row d of (3) contains exactly the prefix $S_1, \dots, S_{61-d}$ together with F when $d > 1$, and column c contains the prefix $S_1, \dots, S_c$ together with F when $c < 60$. Hence with prefix scans $m_j = \min_{i \leq j} S_i$ and $M_j = \max_{i \leq j} S_i$,*

$$\max_d \min_c M(x)_{d,c} = \max_d \begin{cases} m_{60}, & d = 1, \\ \min(m_{61-d}, F), & d > 1, \end{cases}$$

$$\min_c \max_d M(x)_{d,c} = \min_c \begin{cases} \max(M_c, F), & c < 60, \\ M_{60}, & c = 60, \end{cases}$$

computable in $O(60)$ operations, exactly (min and max are associative and commutative on floats), from the 61 class values alone.

Figure 2 draws the layout. Each row contains a prefix of the S array, never an arbitrary subset. Two linear scans therefore reproduce every row minimum and column maximum without materializing the matrix.

4 The per-player dead-TTD quotient

The complete game reads a player’s TTD in exactly one expression: the revival probability (2) of a failed check. That single observation powers the entire reduction.

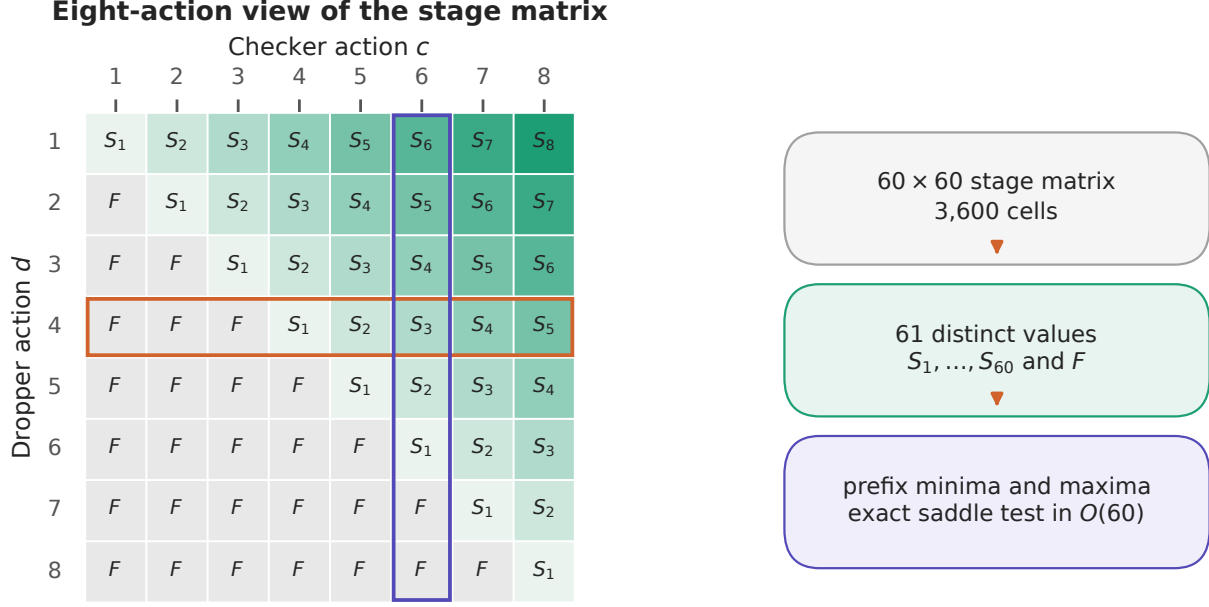


Figure 2: An 8×8 view of the stage matrix (3). The lag $\ell = c - d + 1$ is constant on diagonals, while every failed check has value F . Row minima and column maxima reduce to prefix scans (Lemma 3.4), giving the same result as a scan of all 3,600 cells.

Lemma 4.1 (Death is absorbing). *If the profile (s, t) is dead, then (s', t) is dead for every $s' \geq s$. Since a successful check raises a player's load and leaves their TTD unchanged, and a failed check either ends the game for them or requires (1), load growth is the only motion available to a dead profile, and it never revives it.*

Theorem 4.2 (Quotient soundness). *Define $\varphi(s, t) = (s, t)$ when (s, t) is alive and $\varphi(s, t) = (s, \perp)$ otherwise, where $\perp$ is the dead-TTD sentinel at load s . Apply this map to each player's profile independently. Then any two live states with equal images under φ have identical stage matrices up to φ -identification of their children, and hence identical values: V factors through φ .*

Proof. Compare the 61 classes of Proposition 3.3 at two states that differ only in a dead player's TTD. Success classes never read TTD and preserve both profiles' aliveness classification by Lemma 4.1, so they map to φ -identified children. The failure class reads the checker's TTD only through p : if the checker profile is dead, $p \equiv 0$ and $F = +1$ with no child at all; if alive, the TTD is not quotiented and F is read off unchanged. Induction over the finite acyclic graph (Corollary 3.2) closes the argument. $\square$

Proposition 4.3 (Counting). *Over the transition-closed TTD domain $\{0\} \cup [60, 300]$ (Lemma 4.4), a player has $300 \times 242 = 72,600$ profiles; φ collapses them to 16,711 alive profiles plus 300 dead sentinels (one per load), i.e. 17,011 classes per player and*

$$17,011^2 = 289,374,121$$

two-player classes: a factor 18.2 below the $72,600^2 \approx 5.27 \times 10^9$ reachable states. The raw coordinate envelope $300 \cdot 301 \cdot 300 \cdot 301 \approx 8.15 \times 10^9$, on which the previously recorded closure projection was priced, also counts alive TTDs in 1..59 that Lemma 4.4 rules out; against that envelope the combined restriction and quotient is a factor 28.2.

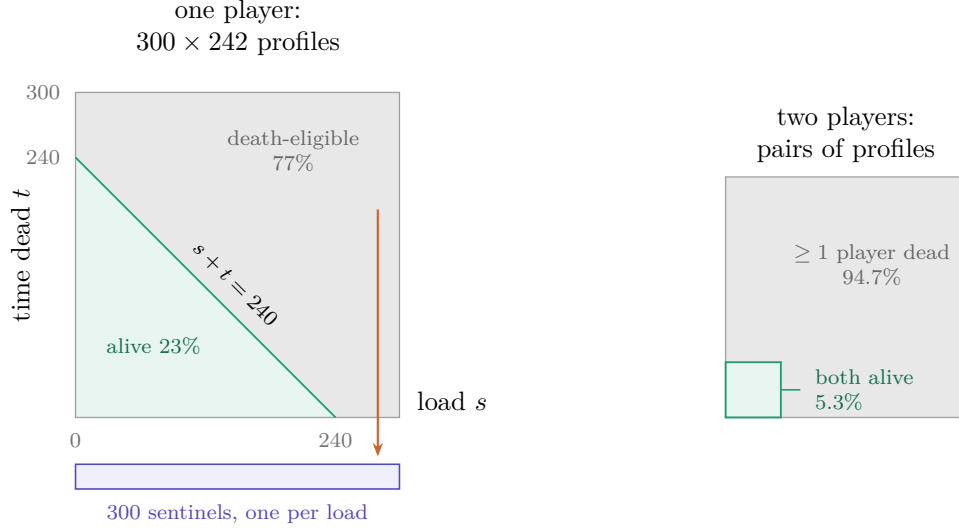


Figure 3: The geometry of the quotient. Left: the alive triangle contains 16,711 of 72,600 profiles; at each load, all dead TTD values map to one of 300 sentinels, leaving 17,011 classes. Right: applying the map to both players reduces 5.27×10^9 states to 289,374,121, a factor of 18.2.

Lemma 4.4 (Domain closure). *Every root of interest has both TTDs in $\{0\} \cup [60, 300]$, and the domain is transition closed: revival requires $s_c + t_c \leq 240$, so a failure child's TTD $t_c + s_c + 60$ lies in $[60, 300]$, and success preserves TTDs. A live state with an alive TTD in 1..59 is thus unreachable from the domain; the implementation rejects such states rather than approximate them. A dead TTD is discarded by φ exactly, so any value is accepted there.*

Figure 3 shows where the factor lives. Death eligibility sounds like a rare endgame condition, but alive profiles occupy only 23% of the full profile rectangle. Applied independently to both players, the quotient leaves only 5.3% of unquotiented state pairs entirely untouched. The survival condition, zero plateau, absorption, and counts were checked by exhaustive enumeration over all 72,600 profiles.

5 The potential and the sweep schedule

Definition 5.1 (Potential). *For a profile let $\rho(s, t) = t$ if (s, t) is alive and $\rho(s, t) = 301$ otherwise; for a class let*

$$\Phi(x) = s_c + s_d + \rho(s_c, t_c) + \rho(s_d, t_d) \in \{0, \dots, 1200\}.$$

Theorem 5.2 (Strict increase). *Every live transition strictly increases Φ . By cases on the moving profile:*

- *successful lag ℓ , profile stays alive or stays dead: $\Delta\Phi = \ell \geq 1$;*
- *successful lag ℓ , alive profile dies: $\Delta\Phi = \ell + 301 - t \geq \ell + 61$, since alive implies $t \leq 240$ (and a dead profile never revives, so the reverse case is empty);*
- *failure, revived profile alive: $\Delta\Phi = 60$ (the load resets to zero while TTD absorbs the dose);*
- *failure, revived profile dead: $\Delta\Phi = 301 - (s_c + t_c) \geq 61$ by (1).*

Corollary 5.3 (Layered backward induction). *No transition stays within a Φ -layer, so a descending sweep solves every child before its parents. Any layer- P residual work need only complete before layer $P - 1$ begins. The 289,374,121 classes occupy all 1,201 layers; the largest, $\Phi = 374$, holds 1,678,715 classes. A layer is a union of bucket rectangles: grouping profiles by their own potential $s + \rho \in [0, 600]$, layer P is exactly $\bigcup_a \text{bucket}(a) \times \text{bucket}(P - a)$.*

The strict increase was checked over all 1,035,541 live profile transitions. The schedule needs no reachability analysis, frontier queue, or transposition table; it simply solves unreachable classes as well.

6 Certified equilibria

Definition 6.1 (Certificate). *For a stage matrix M and mixed strategies σ_d (Dropper) and σ_c (Checker), let*

$$L = \min_c (\sigma_d^\top M)_c, \quad U = \max_d (M \sigma_c)_d, \quad \text{gap}_M(\sigma_d, \sigma_c) = \max(0, U - L).$$

The pair certifies the value $\hat{v} = (L + U)/2$ at tolerance τ_0 when $\text{gap}_M(\sigma_d, \sigma_c) \leq \tau_0$. Throughout, $\tau_0 = 10^{-6}$, and the gap is always measured against the full matrix (3).

Proposition 6.2 (Certified midpoint). *$L \leq \text{val}(M) \leq U$, so a certificate at tolerance τ_0 pins $|\hat{v} - \text{val}(M)| \leq \tau_0/2$.*

Proof. σ_d guarantees the row player at least L against every reply, and σ_c caps the row player at U ; $\text{val}(M)$ sits between the two levels. $\square$

Lemma 6.3 (Matrix-value Lipschitz bound). *If $\|M - \widehat{M}\|_\infty \leq \epsilon$ entrywise, then $|\text{val}(M) - \text{val}(\widehat{M})| \leq \epsilon$, since $|\sigma_d^\top (M - \widehat{M}) \sigma_c| \leq \epsilon$ for all mixed pairs.*

6.1 The solver ladder

Each class is routed through a fixed ladder of certified solvers; the certificate, not the solver, is the acceptance test, so a rung may fail closed for any reason without affecting soundness.

1. **Pure saddle.** Lemma 3.4 computes the maximin and minimax of pure strategies in $O(60)$; if they meet within τ_0 the midpoint is certified by the corresponding pure pair.
2. **Equalizer supports.** Let D be a guessed set of Dropper actions and C a guessed set of Checker actions. Trim both to the same size $k = \min(|D|, |C|)$, then solve

$$\begin{pmatrix} M_{D,C} & -\mathbf{1} \\ \mathbf{1}^\top & 0 \end{pmatrix} \begin{pmatrix} \sigma_{c,C} \\ u_c \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ 1 \end{pmatrix}, \quad \begin{pmatrix} M_{D,C}^\top & -\mathbf{1} \\ \mathbf{1}^\top & 0 \end{pmatrix} \begin{pmatrix} \sigma_{d,D} \\ u_d \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ 1 \end{pmatrix},$$

The first system chooses the Checker's mix so every supported Dropper action earns the same candidate value u_c ; the transpose chooses the Dropper's mix so every supported Checker action yields the same candidate value u_d . For a valid support these values agree. Reject on singularity or negative mass, and certify the lifted pair against the full matrix in $O(60k)$ via (3). Guessed supports come from spatial neighbours one potential step up (the lag-one shift of either profile), recorded when the neighbour was solved. The case $k = 60$ is the classical full-support structured solve and needs no guess. This case matters: 99.8% of the game's equilibria put mass on all 60 actions.

3. **Linear programming.** The standard LP for the row player with the column strategy extracted from its inequality duals, falling back to two independent LPs, and finally to the same two LPs under tightened solver feasibility tolerances (10^{-10} , dual simplex). The retry tightens the *solver*, never the gate: every answer is certified at τ_0 against the full matrix, and a matrix failing all rungs aborts the computation rather than store an uncertified value.

6.2 Propagated error

The table stores certified midpoints, so a parent’s matrix is assembled from child values that are themselves within a bounded distance of truth.

Proposition 6.4 (A priori accumulated bound). *Let every stored value carry a certificate at tolerance τ_0 and let $\delta(x)$ bound the longest live path from x to a terminal. Then the stored value $\widehat{V}(x)$ satisfies*

$$|\widehat{V}(x) - V(x)| \leq \delta(x) \frac{\tau_0}{2} \leq \frac{1200 - \Phi(x)}{2} \tau_0.$$

In particular the initial state obeys $|\widehat{V} - V|(0, 0, 0, 0) \leq 6.0 \times 10^{-4}$.

Proof. Induction up the DAG. Terminal entries are exact. If every child satisfies the bound with $\delta(x) - 1$, the assembled matrix is entrywise within $(\delta(x) - 1)\tau_0/2$ of the true one (expectations and negation are non-expansive), Lemma 6.3 transfers that to the value, and Proposition 6.2 adds the state’s own $\tau_0/2$. Each live transition raises Φ by at least one (Theorem 5.2), bounding δ . $\square$

This a priori bound is loose by design; Section 7 reports the measured gaps, which are typically at the 10^{-12} scale and never exceeded 4.9×10^{-7} on any audited sample. Every stored value’s certificate is re-derivable on demand from the stored children (Bellman recertification), so no per-class certificate is persisted.

7 The computation

7.1 Sweep schedule

Classes are addressed as $p_c \cdot 17,011 + p_d$ and solved one potential layer at a time, with checkpoints between layers. Revival probabilities (2) are evaluated once; the numerical kernels then gather from those values and perform no transcendental arithmetic.

7.2 Cross-implementation audit

The sweep is implemented twice: a Python reference that defines the semantics, and a parallel Rust kernel constrained to *bit-identical* arithmetic: pinned elimination order, first-maximum pivoting, no fused multiply-add, sequential ascending reductions, and per-state warm-start guesses precomputed so that no kernel result depends on thread count or work partition. On every region computed by both (all synthetic end-to-end sweeps, and a forked continuation of the production run) the two implementations produced byte-identical value and routing arrays. The LP residue rung is shared Python code by construction. This does not make the two implementations fully independent, but it shows that the Rust path matches the reference where compared. The complete table was produced once, through the Rust kernel, and every accepted value passed the same certificate gate.

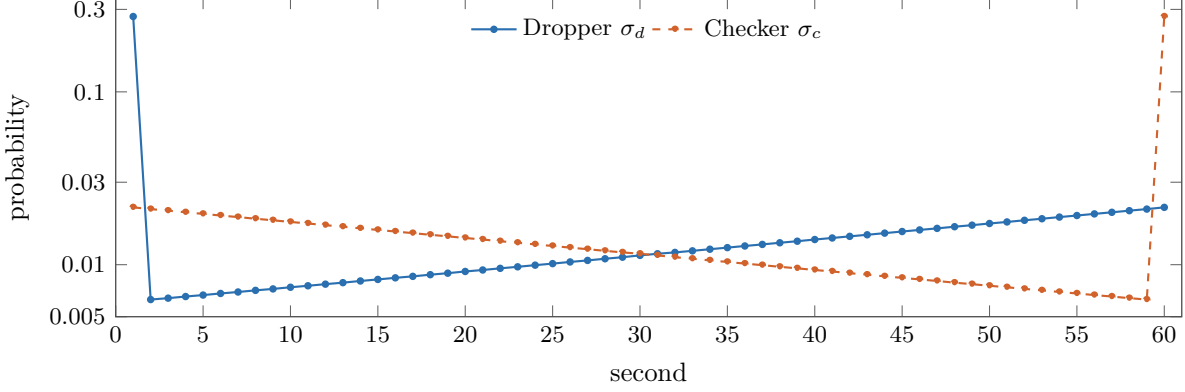


Figure 4: The certified equilibrium at the initial state $(0, 0, 0, 0)$, recertified at gap 1.4×10^{-15} . Both strategies are full-support. The Dropper concentrates 27.3% on dropping immediately and shades the remainder toward later seconds; the Checker mirrors it. The logarithmic axis keeps the atom and the remaining support visible together.

7.3 Results

The full sweep over all 289,374,121 classes completed in 5,432 seconds (1.51 hours) on a single twelve-core consumer desktop (AMD Ryzen 9 3900X), against the previously recorded single-core projection of 3.7–5.1 years for the unquotiented, search-based design. Solver routing:

rung	classes	share
pure saddle (Lemma 3.4)	334,588	0.116%
equalizer, warm-started support	38,138	0.013%
equalizer, full support ($k = 60$)	288,810,400	99.805%
linear programming	190,995	0.066%
of which dual-extraction single LP	190,986	
of which tightened-tolerance retry	9	

Complete-game DTH equilibria are overwhelmingly full-support. This reverses the small-support pattern previously measured on shallow finite-horizon matrices and makes the classical structured solve the workhorse.

Audits on the finished table include a completeness scan (every value finite, in $[-1, 1]$, with a known routing byte); Bellman recertification of 4,792 deterministically sampled classes with worst deviation 4.84×10^{-7} ; agreement with the independently produced endgame band tablebase (both loads ≥ 240) on all 3,541 shared classes to at most 4.9×10^{-8} , and at its root to 6.9×10^{-12} ; and an independent dead-band reference solver sharing no sweep machinery.

Theorem 7.1 (Main result). *The complete Drop-the-Handkerchief game of Section 2, restricted to the transition-closed quotient domain of Lemma 4.4, is solved: every one of the 289,374,121 classes carries a stored value certified at saddle-gap tolerance 10^{-6} , with the a priori accumulated bound of Proposition 6.4 and the measured audits above. At the initial state,*

$$\widehat{V}(0, 0, 0, 0) = 0.0898500728095116,$$

Thus under optimal play the first Dropper wins with probability $\frac{1+\widehat{V}}{2} \approx 0.5449$, within the stated error bounds; equilibrium strategies are re-derived from stored child values on demand.

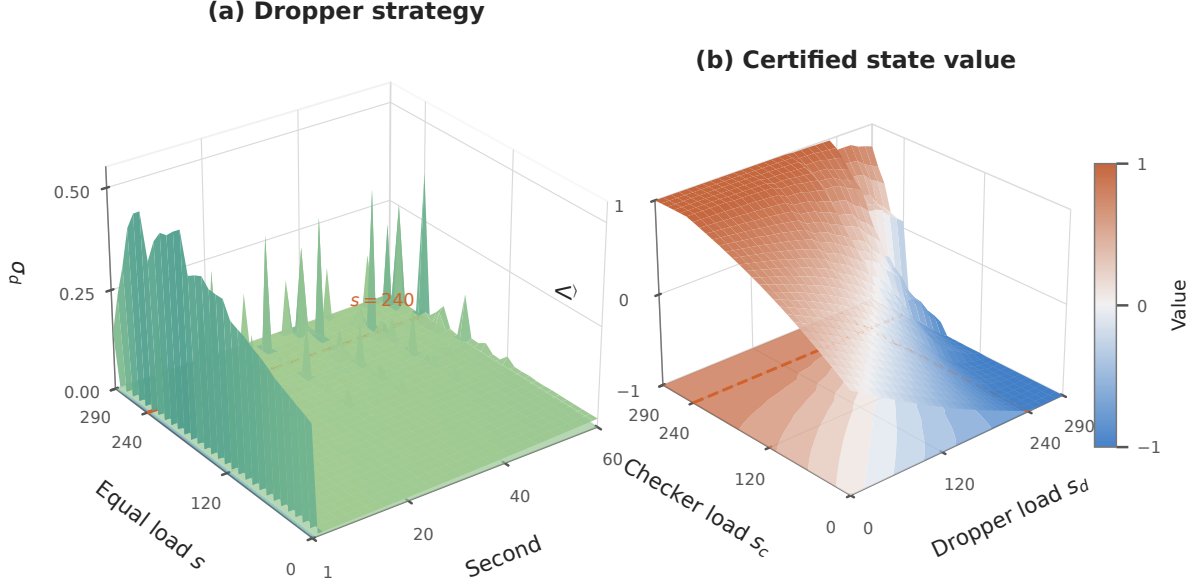


Figure 5: Two zero-TTD slices, sampled in 10-load steps. **(a)** The Dropper’s equilibrium mix along $(s, 0, s, 0)$. Every sampled strategy has full support; the survival boundary at $s = 240$ produces atoms on seconds 1 and 60, before the mix flattens near overflow. **(b)** Certified values $\hat{V}(s_c, 0, s_d, 0)$. At equal loads of 290, the mover’s value is 0.753; unequal high loads create the opposing cliffs.

8 Scope of the claims

- **Rules-relative.** The theorems are about the game defined in Section 2 with the frozen revival surface (2). Nothing here solves the leap-aware “Surpassing the Leader” game, which adds an asymmetric 61st second and is a different game.
- **Certified numerics, not rational arithmetic.** Values are IEEE double midpoints with explicit saddle-gap certificates and the accumulated bound of Proposition 6.4; this is a certified-numerics result, not an exact rational or interval-arithmetic proof. The structural theorems (quotient, potential, finiteness, Toeplitz form) are exact and machine-checked exhaustively.
- **Domain.** The table covers the transition-closed quotient domain; live states with an alive TTD in 1..59 are outside it and are rejected, not approximated (Lemma 4.4).
- **Values, not tabulated policies.** The table stores values and routing only; policies are re-derived and re-certified per query, as in Figures 4 and 5.

Reproducibility

From the repository head: `uv run python -m dth complete` builds the artifact (checkpointed, resumable; the configuration is `src/dth/config/complete_full_v1.yaml`); the exhaustive structural tests and both-backend parity suite run under `uv run python -m pytest src/dth/tests -q` and `cargo test --workspace`. The artifact manifest embeds SHA-256 digests of every array and of the solver sources that produced it.

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